Solution Design — FoxOtale (Guest App + Staff App)

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Stakeholders: Product / Design, Engineering, Legal / Finance, Hotel owners / partner properties
Status: Approved (Phase 1B)
Active delivery phase: Phase 1 (MVP) — Phases 2–3 are roadmap only
Platform: Flutter (iOS + Android) — Guest App and Staff App
Source: `foxOtale/FoxOtale-Guest-App-PRD.docx`, `foxOtale/FoxOtale-Staff-App-PRD.docx`

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1. Business Objectives

- **Primary goal of this feature:**
  Ship paired Flutter mobile apps so FoxOtale partner hotels can take direct bookings (Guest App) and run property setup, daily operations, and management intelligence from one role-aware Staff App — reducing OTA dependence and tool sprawl.

- **Key business outcomes / KPIs this drives:**
  - Increase **direct-booking share** vs OTA for launch properties (primary lagging metric)
  - Guest **checkout completion ≥ 70%**; **time-to-book < 4 min** for returning guests
  - Guest App becomes the **system of record** for vouchers, bills, and final/GST invoices
  - Staff: cut median **guest-request → action** time; raise **task completion on shift** and **owner engagement** with live dashboards/reports

- **Alignment to company priorities / OKRs:**
  [ASSUMPTION: maps to FoxOtale brand promise “Make every stay a story” and ownership of the direct guest relationship.] This Solution Design covers **Phase 1 (MVP)** only; Phases 2–3 remain roadmap.

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2. Problem Statement / User Needs

- **Current pain points / inefficiencies:**
  - Independent/boutique hotels lose **15–25%** of every booking to OTA commissions and retain almost no post-checkout guest relationship.
  - Guests juggle multiple apps to discover, book, and contact the property; there is no branded place for the full stay and financial documents.
  - Hotel teams run ops across PMS, spreadsheets, WhatsApp, phone, and paper — owners lack a live operating picture; tasks and guest requests get lost between roles.

- **Who is impacted (user personas / roles):**

Persona / role	Impact
Leisure traveller	Needs inspiration + confidence the property is right
Corporate / business traveller	Needs speed, GST/invoices, saved preferences
Last-minute booker	Needs fast search, live availability, quick pay
Returning guest	Needs rebook path and recognition (full rebook UX is Phase 2)
Owner	Needs self-serve property setup + revenue/occupancy visibility (incl. chain)
Area GM / Area HM (chain)	Needs city/destination views across hotels
GM / Hotel Manager	Needs full property ops + reports + setup
Ops Manager	Needs task assignment, rooms/inventory, request resolution
Supervisor	Needs shift queue and room/task status updates
Executive / Frontline	Needs own-task clarity and completion

- **Evidence / data supporting the need:**
  [ASSUMPTION: based on stakeholder PRDs — OTA commission range 15–25% and ops tool sprawl described qualitatively; no baseline conversion/response-time telemetry attached.]

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3. Scope

In Scope (Phase 1 MVP):

Guest App

- Onboarding & auth (phone OTP + social/email); browse before login; login required at checkout
- Home / discovery, search + live availability, filters & sort, property detail
- Room selection, checkout & payment (UPI/cards/wallets; GST/company field), booking confirmation
- Confirmation voucher (view / download / share)
- Guest dashboard: Past / In-house / Upcoming; documents per booking; monthly view
- Final bill / GST invoice after check-out; booking modify/cancel within policy; profile

Staff App

- Role-based auth & strict hierarchy (Owner → Area GM/HM → GM/HM → Ops Mgr → Supervisor → Executive); one role per person in v1
- Role-aware dashboards; standalone + chain (hotel-wise / city-destination switch)
- Property setup / onboarding (rooms, rates, meal plans, inventory, facilities, nearby & distances)
- Reservation dashboard (arrivals, departures, in-house, revenue, pay received vs pending)
- Bookings view; room & inventory status; task management; guest-requests queue (contract ready; volume ramps with Guest Phase 2); notifications; profile

- Intelligence (manual data entry in v1): core reports (YoY/MoM/budget vs actual), segment/source productivity, revenue & ARR, revenue module (rates/meal plans/blackouts/high-occupancy), guest management (revenue split), corporate/TA/event-company data

Out of Scope (for this phase):

- Guest: multi-brand/OTA aggregation; flights/cabs/full trip planning; loyalty points; social feed; responsive web booking parity
- Staff: full PMS replacement; accounting/payroll; channel manager / rate distribution; deep HR/rostering; automated forecasting / dynamic pricing AI
- `[ROADMAP — Guest Phase 2]` Digital check-in, in-stay services, chat, reviews, add-ons, one-tap rebook, wishlist, offers
- `[ROADMAP — Guest Phase 3]` Loyalty, multi-property trips, experiences marketplace, referrals, multi-language/currency
- `[ROADMAP — Staff Phase 2]` Team management, approvals/escalations, guest chat, task photo proof, long-stay list, demand calendar
- `[ROADMAP — Staff Phase 3]` Rostering/leave, deep OTA/PMS auto-feeds, multi-role users, staff KPI scoring, channel manager, dynamic pricing

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4. User Stories / Scenarios

> Priority: **Must** / **Should** / **Could** / **Won't**
> Jira keys backfilled in Phase 3.

Guest App

- **US-G01 (Must):** As a last-minute booker, I want to search by location, dates, and guests and see only available properties with live price so that I do not waste time on sold-out options. — **Jira:** *pending*
- **US-G02 (Must):** As a guest, I want a rich property page (photos, rooms, amenities, map, reviews, all-in price) so that I can decide with confidence. — **Jira:** *pending*
- **US-G03 (Must):** As a guest, I want to complete checkout with UPI/cards/wallets and receive instant confirmation so that my booking is secure. — **Jira:** *pending*
- **US-G04 (Must):** As a guest, I want a confirmation voucher I can view, download, and share so that I have proof of booking. — **Jira:** *pending*
- **US-G05 (Must):** As a guest, I want a dashboard of bookings (Past / In-house / Upcoming) with documents so that vouchers, bills, and invoices live in one place. — **Jira:** *pending*
- **US-G06 (Must):** As a corporate traveller, I want GST/company details at booking and a GST invoice after check-out so that I can expense the trip. — **Jira:** *pending*
- **US-G07 (Should):** As a leisure traveller, I want to browse featured and nearby properties on home so that I get inspired before I have fixed plans. — **Jira:** *pending*
- **US-G08 (Should):** As a guest, I want filters (price, rating, amenities, type) and sort so that I can narrow results quickly. — **Jira:** *pending*
- **US-G09 (Should):** As a frequent traveller, I want bookings and documents grouped by month so that I can reconcile expenses. — **Jira:** *pending*
- **US-G10 (Should):** As a guest, I want to modify or cancel within published policy and see refund status so that plans can change. — **Jira:** *pending*

- **US-G11 (Could):** As a guest, I want to browse before login so that discovery is not blocked by signup. — **Jira:** *pending*
- **US-G12 (Won't — Phase 2):** As a guest, I want digital check-in, in-stay requests, and chat with the property so that I skip the front desk and phone. — **Jira:** *pending*

Staff App

- **US-S01 (Must):** As an owner/GM, I want to set up a property (rooms, rates, meal plans, inventory, facilities, nearby & distances) so that the hotel becomes bookable without external help. — **Jira:** *pending*
- **US-S02 (Must):** As any staff member, I want role-based login that shows only my permitted screens/actions so that access matches the hierarchy. — **Jira:** *pending*
- **US-S03 (Must):** As a manager, I want a reservation dashboard (arrivals, departures, in-house, revenue, paid vs pending) so that I know today's position at a glance. — **Jira:** *pending*
- **US-S04 (Must):** As an ops manager/supervisor, I want to manage room status and assign/complete tasks so that work is owned and availability is accurate. — **Jira:** *pending*
- **US-S05 (Must):** As an owner with a chain, I want hotel-wise and city/destination consolidated views so that I can compare and drill down. — **Jira:** *pending*
- **US-S06 (Should):** As a manager, I want YoY/MoM and budget-vs-actual reports (today/MTD/YTD) with export so that I track performance without spreadsheets. — **Jira:** *pending*
- **US-S07 (Should):** As a revenue manager, I want to manage rates, meal plans, black-out/sold-out, and high-occupancy dates so that pricing reflects demand. — **Jira:** *pending*
- **US-S08 (Should):** As a manager, I want per-guest and corporate/TA/event revenue views so that I can prioritise relationships. — **Jira:** *pending*
- **US-S09 (Should):** As an ops user, I want a guest-requests queue fed from the Guest App so that requests are tracked to resolution. — **Jira:** *pending*
- **US-S10 (Could):** As any staff member, I want push notifications for new tasks/requests/escalations so that I respond fast. — **Jira:** *pending*
- **US-S11 (Won't — Phase 2):** As a GM, I want structured approvals, team management, and guest chat so that control and communication stay in-app. — **Jira:** *pending*

High-level user journeys (Phase 1):

Guest — Discover → Book → Record

1. Open app → browse home or search (location + dates + guests)
2. Open property → evaluate rooms/price → select room
3. Checkout (auth if needed) → pay → confirmation + voucher
4. Dashboard shows Upcoming booking + documents
5. After check-out → final bill / GST invoice appears on booking

Staff — Setup → Run → See numbers

1. Owner/GM completes property setup wizard → property publishable/bookable
2. Role login → role-scoped dashboard (chain switch if Owner/Area)
3. Use reservation dashboard + bookings + room board + tasks for the day
4. Enter/manage report, revenue, guest, and corporate data; view/export reports
5. Guest-requests queue ready for assigned resolution (traffic grows with Guest Phase 2)

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5. Functional Requirements

Requirement ID	Description	Priority	Jira Epic	Acceptance Criteria
FR-G001	Guest onboarding & auth	Must	*pending*	Phone OTP + ≥1 social/email; browse pre-login; login enforced at checkout
FR-G002	Search + availability + live price	Must	*pending*	Search by location, date range, guests; results only bookable inventory with live price
FR-G003	Property detail	Must	*pending*	Gallery, rooms, amenities, map/location, reviews, all-in price with taxes
FR-G004	Room selection & booking draft	Must	*pending*	Choose room, dates, guest count; price breakdown incl. taxes before pay
FR-G005	Checkout & payment	Must	*pending*	UPI/cards/wallet; GST/company field; secure pay; failure retry/messaging
FR-G006	Booking confirmation	Must	*pending*	Instant confirmation UI + notification/email with booking ID and details
FR-G007	Confirmation voucher	Must	*pending*	PDF voucher with ID, property, dates, room, guest, amount; view/download/share; also via email/push
FR-G008	Guest dashboard — bookings	Must	*pending*	Past / In-house / Upcoming segments with status + detail; default for signed-in users
FR-G009	Final bill / GST invoice	Must	*pending*	After check-out, final bill generated/shared; GST invoice when GST details provided; stored on booking
FR-G010	Home / discovery	Should	*pending*	Featured, nearby, and search entry on first screen
FR-G011	Filters & sort	Should	*pending*	Filter price/rating/amenities/type; sort price/rating/distance
FR-G012	Dashboard — documents + monthly view	Should	*pending*	Per-booking voucher/bill/invoice; download/share; toggle monthly grouping

Requirement ID	Description	Priority	Jira Epic	Acceptance Criteria
FR-G013	Booking modify/cancel	Should	*pending*	Within policy; policy shown; refund status visible
FR-G014	Guest profile	Should	*pending*	Personal details, saved GST, payment methods, preferences
FR-G015	Pay-at-hotel path	Could	*pending*	[ASSUMPTION: allowed only where property flag enables it] Shown only when property allows
FR-G016	In-stay services / digital check-in / chat / reviews	Won't	*pending*	Explicitly Phase 2
FR-S001	Role-based auth & access control	Must	*pending*	Login resolves one role; UI hides unauthorised actions per §7 matrix (draft)
FR-S002	Role-aware dashboards	Must	*pending*	Each role lands on scoped dashboard (owner financial; supervisor ops queue; etc.)
FR-S003	Property setup / onboarding	Must	*pending*	Rooms, rates, meal plans, inventory, facilities, nearby & distances; bookable when complete
FR-S004	Reservation dashboard	Must	*pending*	Arrivals, departures, in-house, revenue, payment received vs pending for selected property/date
FR-S005	Bookings view	Must	*pending*	Role-appropriate list/calendar + detail; create/modify/cancel per matrix
FR-S006	Room & inventory status	Must	*pending*	Live room states; authorised updates reflect app-wide
FR-S007	Task management	Must	*pending*	Create/assign (authorised); execute/complete (assignee); status + owner on every task
FR-S008	Standalone & chain support	Must	*pending*	Single hotel works; Owner/Area can switch hotel-wise and city/destination-wise; entities scoped by property
FR-S009	Guest requests queue	Should	*pending*	Ingest Guest App requests; assign; track to resolution (contract in P1 even if volume low)

Requirement ID	Description	Priority	Jira Epic	Acceptance Criteria
FR-S010	Notifications	Should	*pending*	Role-relevant push for tasks, requests, escalations
FR-S011	Reports — core (YoY/MoM/budget vs actual)	Should	*pending*	Today/MTD/YTD; export; data entered/managed in-app for v1
FR-S012	Segment & market-source productivity	Should	*pending*	Room nights + revenue productivity daily/monthly/custom; in-app entry
FR-S013	Revenue & ARR report (budget vs actual)	Should	*pending*	Daily/monthly/custom; in-app entry
FR-S014	Revenue module (rates, meal plans, blackouts, high-occupancy)	Should	*pending*	Authorised roles manage pricing controls used by availability
FR-S015	Guest management analytics	Should	*pending*	Month-wise & alphabetical; per-guest revenue split room/F&B/banquet/extras per hotel
FR-S016	Corporate / TA / event company data	Should	*pending*	Alphabetical; month & year; ranked by room nights/ARR/revenue
FR-S017	Staff profile & settings	Could	*pending*	Everyone manages own profile
FR-S018	Team mgmt / approvals / guest chat / demand calendar	Won't	*pending*	Explicitly Phase 2
FR-S019	OTA/PMS auto-feed of intelligence data	Won't	*pending*	Explicitly Phase 3; v1 is manual entry

Priority mix: 17 Must / 14 Should / 3 Could / 4 Won't among detailed FRs above (~45% Must of Must+Should+Could actionable set) — within the ≤60% Must guideline.

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6. UI/UX Requirements

- **Layout / responsiveness:**
  Mobile-first Flutter apps (iOS + Android). Guest App: expressive, brand-forward consumer UX. Staff App: calmer, denser, high-clarity ops UX. Support phones as primary; tablet layouts nice-to-have (Could).
- **Key interactions:**
  - Guest: search sheet (location/dates/guests) → results list/map toggle → property detail → room select → checkout → payment result → voucher actions (view/share) → dashboard tabs
  - Guest: documents attached to booking detail; monthly toggle regroups list

    - Staff: setup wizard steps with progress; chain switcher (Owner/Area); reservation day strip; room board status chips; task accept/complete; report date range pickers + export
    - Staff: unauthorised modules never shown (not merely disabled)
    - Empty / error / offline-tolerant states for weak-signal staff updates [ASSUMPTION: offline queue for status patches required for frontline]

- **Design component mapping (Flutter — not Epsilon CORE UI):**
  FoxOtale is a **Flutter** product. Epsilon **CORE UI** (`@epsilon/core-ui`) is Angular/web-oriented and is **not** the implementation target. Map to Flutter / Material 3 (and iOS Cupertino where platform-adaptive) plus a FoxOtale brand theme:

Need	Flutter / Material mapping
Primary actions	`FilledButton` / `ElevatedButton`
Secondary / text actions	`OutlinedButton` / `TextButton`
Forms	`TextFormField`, `DropdownButtonFormField`, date/range pickers
Lists & boards	`ListView` / `CustomScrollView`; Staff room board as grid of status cards
Tabs (dashboard segments)	`TabBar` + `TabBarView` (Past / In-house / Upcoming)
Filters	Bottom sheet / `FilterChip` / `ChoiceChip`
Status	Custom `Chip` / badge widgets with semantic colors
Navigation	Guest: bottom `NavigationBar`; Staff: role-scoped drawer or bottom nav
Confirmations	`showDialog` / modal bottom sheets
Charts (reports)	`fl_chart` or equivalent [CUSTOM — pick in Phase 2]
PDF voucher/invoice view	`pdf` + share via `share_plus` [CUSTOM]
Map on property	Google Maps / Mapbox Flutter plugin [CUSTOM — provider TBD]
Payments	Native UPI/card SDK / gateway plugin [CUSTOM — PSP TBD]
Push	FCM + APNs

> Brand: Guest App may be louder on discovery; checkout and Staff data-dense screens prioritise clarity over brand ornamentation. Exact brand system to be defined in Phase 2 UX Ideation.

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7. Non-Functional Requirements

- **Performance:**
  Guest: search results interactive < 2s on typical 4G; payment confirmation perceived as immediate

after PSP success. Staff: reservation dashboard and room board refresh useful within a few seconds of status change. Support concurrent staff on a property during peak check-in/out.

- **Security:**
  Guest: secure auth (OTP/social); PCI-aware payment via PSP (no raw card storage in app). Staff: strict RBAC per hierarchy; hide unauthorised UI; audit log for booking changes, rate changes, and task reassignment [ASSUMPTION: audit log required for management actions]. Transport TLS; secure token storage (Keychain/Keystore).

- **Accessibility:**
  Target WCAG 2.1 AA as practical on mobile: sufficient contrast, scalable text, screen-reader labels on primary flows (book, pay, complete task).

- **Browser / device support:**
  iOS and Android current and previous major versions (Flutter). No web booking engine in this SD (tracked separately).

- **Data / integrations:**
  - Property inventory, rates, availability — source of truth TBD (manual Staff setup vs PMS/channel manager) `[ASSUMPTION]`
  - Payment service provider (UPI/cards/wallets)
  - Notification/email delivery for confirmation + documents
  - Document generation service for voucher / final bill / GST invoice — **single source of truth TBD** (Guest App, Staff App at check-out, or PMS)
  - Guest ↔ Staff request routing contract (Staff queue in P1; Guest producers primarily Phase 2)
  - No codebase scan performed (greenfield; no FoxOtale application code in workspace)

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8. Design System & Implementation Notes

- **Framework:** Flutter (Dart) — Guest App and Staff App (prefer shared packages for auth, booking models, document types, design tokens where safe)

- **Design system:** FoxOtale Flutter theme (Material 3 base + brand tokens). **Not** Epsilon CORE UI.

- **Module split (suggested):**
  - `apps/guest_app`, `apps/staff_app`
  - `packages/foxotale_ui` (tokens, shared widgets)
  - `packages/foxotale_domain` (booking, property, document, task models)
  - `packages/foxotale_api` (clients)

- **Expected Flutter building blocks:**

Area	Building blocks
Navigation	`go_router` (or equivalent)
State	Team-standard (Riverpod / Bloc / Provider — pick in eng kickoff)

Area	Building blocks
Forms	Flutter Form + validators
Lists / dashboards	Material lists, cards, tables (Staff reports)
Charts	Chart package TBD
Documents	PDF generate/view/share
Maps	Maps SDK TBD
Payments	PSP Flutter SDK TBD
Push	`firebase_messaging` (or equivalent)

- **Design tokens to apply:**
  Brand palette, typography, spacing, elevation — Guest more expressive; Staff denser type/spacing. Define in Phase 2.

- **Custom component needs (flag for design/eng):**
  - Property photo gallery + map hybrid
  - Room status board
  - Reservation day dashboard
  - Chain switcher (hotel / city)
  - PDF voucher & GST invoice templates
  - Payment result + failure recovery UI
  - Setup wizard (multi-step property onboarding)
  - Report hub with export

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9. Success Metrics & Acceptance

- **How will we measure success post-launch?**

Metric	Target / note	Horizon
Booking conversion (search → confirmed)	Beat property OTA baseline	Leading
Checkout completion	≥ 70%	Leading
Time to book (returning guest)	< 4 minutes median	Leading
Direct-booking share	Primary business KPI	Lagging
Repeat booking rate	Track uplift vs pre-app	Lagging
Request response time (staff)	Median request → first action	Leading (ramps with Guest Phase 2)
Task completion on shift	% completed	Leading

Metric	Target / note	Horizon
Owner dashboard engagement	Owners use app vs asking for MIS	Lagging

- **Definition of Done:**

Stage	Done When
Solution Design	All 10 sections complete; approved by Product and Engineering Lead
UX Ideation	Screen/component inventory + journeys validated; Flutter widget map agreed
Prototype	Flutter prototype (or click-through) covers critical Guest book path + Staff setup/reservation path
Implementation	All Must and agreed Should FRs implemented; RBAC verified; document generation path agreed and working; a11y spot-check on primary flows
Launch	Deployed to stores (or staged rollout) for pilot properties; metrics baseline established

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10. Risks / Dependencies / Assumptions / Open Questions

Risks:

- **Invoice/voucher source of truth unclear** — dual generation or missing docs at check-out. *Mitigation: decide owner (Staff check-out vs service vs PMS) before build.*
- **Inventory/pricing without PMS** — stale availability → overbooking. *Mitigation: explicit v1 source (manual Staff rates + inventory) and overbooking policy.*
- **Intelligence module size** — manual entry burden may reduce Staff adoption. *Mitigation: minimal entry paths; Phase 3 auto-feeds designed into data model now.*
- **Guest requests queue idle in P1** — Staff P0-6 depends on Guest Phase 2 producers. *Mitigation: ship queue + API contract; seed with manual/staff-created requests if needed.*
- **Permission matrix still draft** — wrong access → security/UX rework. *Mitigation: Product sign-off on §7 before implementation.*

Dependencies:

- Payment service provider selection and Flutter SDK
- Email/SMS/push providers
- Document (PDF) generation + GST template from Legal/Finance
- Agreement on check-out → invoice trigger between Guest and Staff
- Map provider; auth providers (OTP/social)
- No codebase-derived dependencies (scan skipped — greenfield)

Assumptions:

- [ASSUMPTION]: Phase 1 Guest inventory is FoxOtale partner properties only, configured via Staff setup.

- [ASSUMPTION]: One role per staff user in v1.
- [ASSUMPTION]: Intelligence data is manually entered/managed in Staff App for v1.
- [ASSUMPTION]: Staff check-out action is the trigger that produces final bill/GST for the Guest dashboard (pending confirmation).
- [ASSUMPTION]: Pay-at-hotel is property-flagged, not globally always-on.
- [ASSUMPTION]: Chain “area” ≈ city/destination cluster; a property belongs to one area in v1.
- [ASSUMPTION]: Monthly document view is per-guest only in P1 (no multi-employee corporate consolidation).

Open Questions / Decisions Needed:

#	Question	Owner	Target Date	Decision
Q1	Launch properties; inventory/availability real-time from PMS or manual?	Business + Eng	TBD	Pending
Q2	Pay-at-hotel allowed? Which properties?	Business	TBD	Pending
Q3	Source of truth for availability & pricing (channel manager / PMS / Staff App)?	Eng	TBD	Pending
Q4	Who generates confirmation voucher and final/GST invoice?	Eng + Product	TBD	Pending
Q5	Final bill before vs only after check-out; interim bill for in-house?	Business	TBD	Pending
Q6	GST invoice mandatory fields / template	Legal / Finance	TBD	Pending
Q7	Cancellation/refund policy: global vs per-property?	Legal + Business	TBD	Pending
Q8	Guest data at signup vs deferred (privacy)	Product + Legal	TBD	Pending
Q9	Confirm Staff §7 permission matrix (Owner setup, Area roles, intelligence view-only below GM)	Business / Product	TBD	Pending
Q10	Area definition for chains; multi-area membership?	Business	TBD	Pending
Q11	Who enters budgets and at what cadence?	Business / Finance	TBD	Pending
Q12	Offline required for frontline status updates?	Eng + Ops	TBD	Pending

Appendix

- **Related docs / links:**
  - `foxOtale/FoxOtale-Guest-App-PRD.docx` (imported Markdown: `.sd/foxotale/FoxOtale-Guest-App-PRD.md`)

    - `foxOtale/FoxOtale-Staff-App-PRD.docx` (imported Markdown: `.sd/foxotale/FoxOtale-Staff-App-PRD.md`)
    - Staff PRD §7 Role/Permission Matrix — **DRAFT — please correct**

- **Jira Links** *(backfilled in Phase 3):*

Artifact	Jira Key	Notes
Capability (MVP)	*pending*	Parent for MVP epics
Epic E*	*pending*	See epic-breakdown.md
Story US-*	*pending*	

- **Mockups / wireframes:**
    - Wireframes / Flutter screen inventory to be created in Phase 2 (UX Ideation)
- **UX Ideation artifact:**
    - `.sd/foxotale/ux-ideation.md` (pending Phase 2)